

What all can Archimedes model do

- Physics effects and transport for electrons and heavy holes in silicon, germanium, GaAs, InSb, AlSb, AlAs, AlInxSb, AlIn(1-x)Sb, AlP, AlSb, GaP, GaSb, InP and their compounds (III-V semiconductor materials), along with silicon oxide, the applied and/or self-consistent electrostatic and magnetic fields by means of Poisson and Faraday equations.
- Pre-defined device types, such as diodes, metal semiconductor field effect transistors (MESFETs) and metal oxide field effect transistors (MOSFETs), are built in for nanoHUB users, with geometry dimensions, doping levels and numerical simulation controls as adjustable parameters.
- Other user-defined advanced devices.

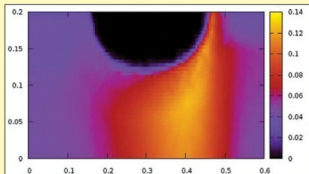


Fig. 3: Simulated electron energy of a MESFET

weighted Bohm potential, full effective potential and density-gradient potential.

The electrostatic potential is described with the help of a well-known Poisson equation, which is sometimes called Hartree approxi-

Hartree approximation for electrostatic potential. The transport problem in semiconductor devices can be split into the equation describing dynamics in a given potential and the equation describing the potential generated by electrons.

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mation for electrons-electron interaction. This description is helpful in modelling the electron transport to a great extent. In future, some phenomenon, like exchange-correlation effect, has to be incorporated into this model.

The first free Monte Carlo simulator

Archimedes is never the last word for semiconductor device simulators. Even developers do not claim so. The relevance of the simulator comes from the fact that this is the first open source simulator of its kind. Although many capabilities are yet to be implemented, developers are optimistic in including these in future versions. Hopefully, the simulator can cater to the needs of a semiconductor scientist, who needs a good starting point for his or her work to share it with the society. ●

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